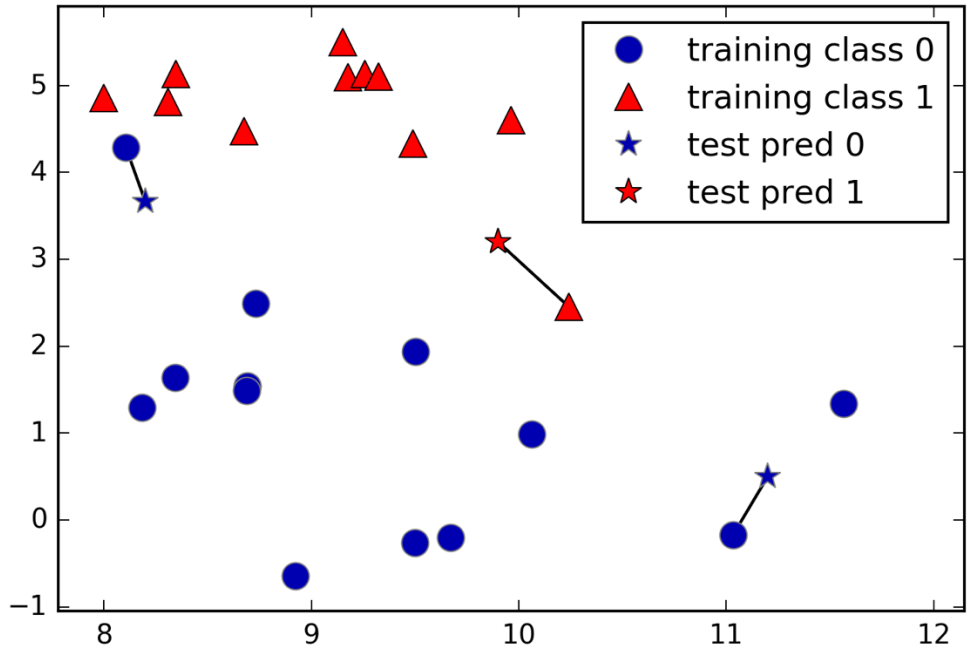
COMP 3250
Data Analytics

Week 3 Supervised Learning

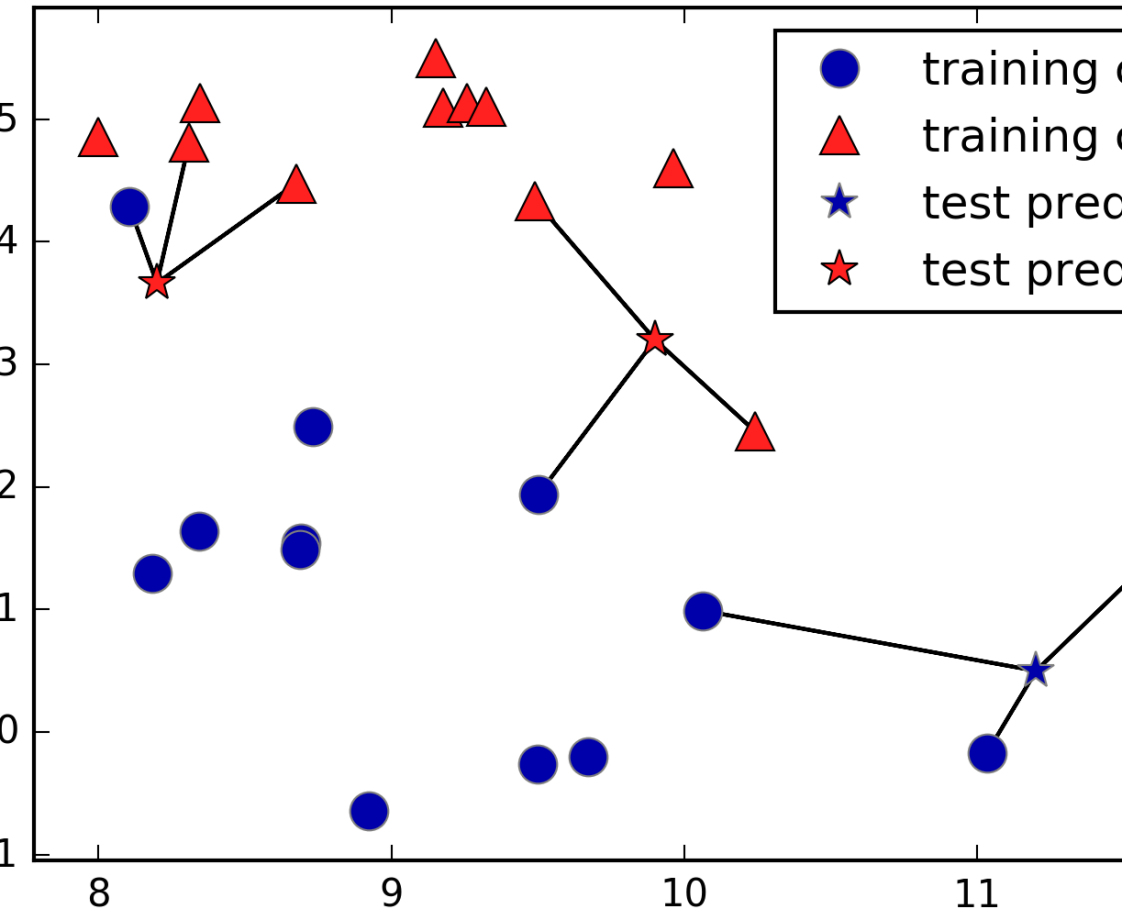
With examples and figures from Introduction to Machine Learning with Python A Guide for Data Scientists By Andreas C. Müller, Sarah Guido · 2016

k-Neighbors classification



Building the model consists only of storing the training dataset. To make a prediction for a new data point, the algorithm finds the closest data points in the training dataset—its “nearest neighbors.”

In its simplest version, the k-NN algorithm only considers exactly one nearest neighbor, which is the closest training data point to the point we want to make a prediction for



Instead of considering only the closest neighbor, we can also consider an arbitrary number, k, of neighbors. This is where the name of the k-nearest neighbors algorithm comes from. When considering more than one neighbor, we use voting to assign a label. This means that for each test point, we count how many neighbors belong to class 0 and how many neighbors belong to class 1. We then assign the class that is more frequent: in other words, the majority class among the k-nearest neighbors

Class examples(kNN)

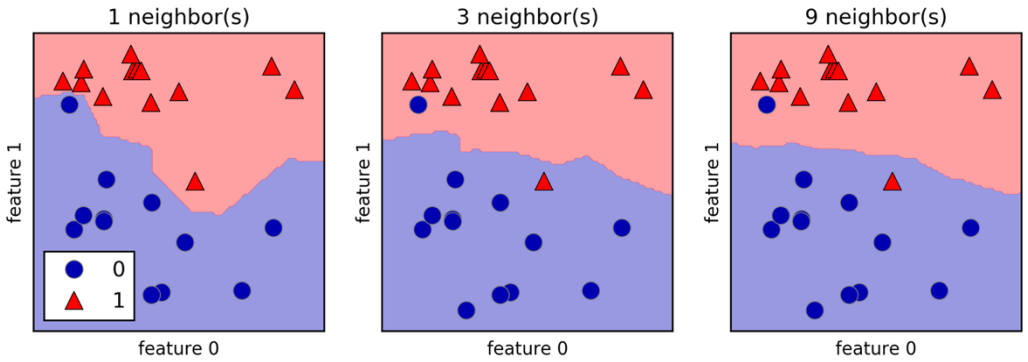
```
fig, axes = plt.subplots(1, 3, figsize=(10, 3))
for n_neighbors, ax in zip([1, 3, 9], axes):
# the fit method returns the object self, so we can instantiate
# and fit in one line
clf = KNeighborsClassifier(n_neighbors=n_neighbors).fit(X, y)
mglearn.plots.plot_2d_separator(clf, X, fill=True, eps=0.5, ax=ax, alpha=.4)
mglearn.discrete_scatter(X[:, 0], X[:, 1], y, ax=ax)
ax.set_title("{} neighbor(s)".format(n_neighbors))
ax.set_xlabel("feature 0")
ax.set_ylabel("feature 1")
axes[0].legend(loc=3)
```

Analyzing KNeighborsClassifier

For two-dimensional datasets, we can also illustrate the prediction for all possible test points in the xy-plane. We color the plane according to the class that would be

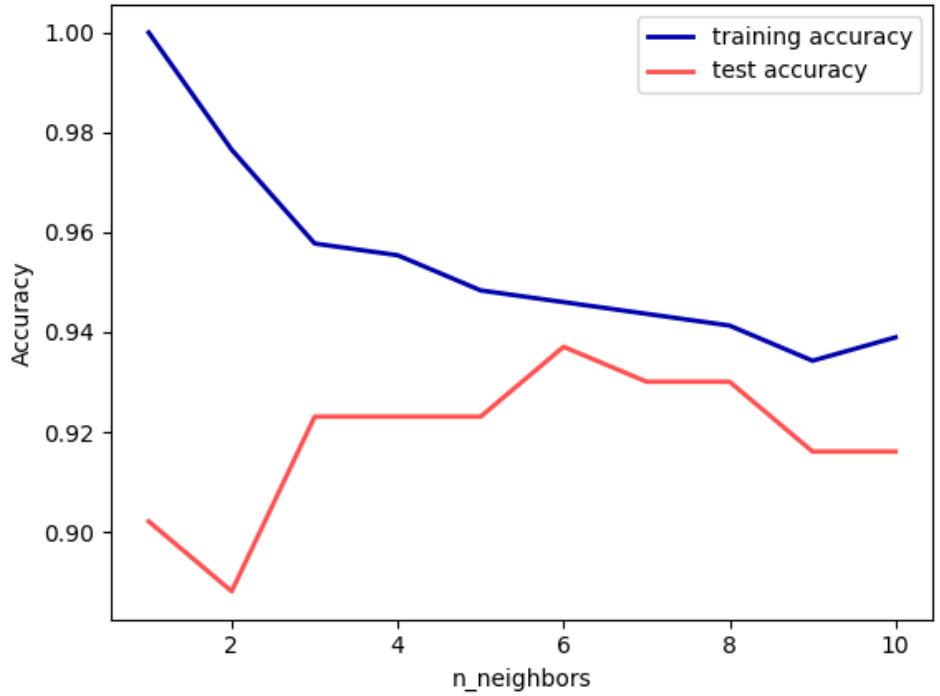
assigned to a point in this region. This lets us view the decision boundary, which is the divide between where the algorithm assigns class 0 versus where it assigns class 1.

Decision boundaries created by the nearest neighbors model for different values of n_neighbors



A smoother boundary corresponds to a simpler model. In other words, using few neighbors corresponds to high model complexity (as shown on the right side of on model complexity presented in previous weeks), and using many neighbors corresponds to low model complexity (as shown on the left side of Figure on Model complexity present in previous weeks). If you consider the extreme case where the number of neighbors is the number of all data points in the training set, each test point would have exactly the same neighbors (all training points) and all predictions would be the same: the class that is most frequent in the training set

The connection between model complexity and generalization using the Breast Cancer Dataset:
We begin by splitting the dataset into a training and a test set. Then, we evaluate training and test set performance with different numbers of neighbors.



Comparison of training and test accuracy as a function of n_neighbors

```
In [41]: from sklearn.datasets import load_breast_cancer

cancer = load_breast_cancer()
X_train, X_test, y_train, y_test = train_test_split(
    cancer.data, cancer.target, stratify=cancer.target, random_state=66)

training_accuracy = []
test_accuracy = []
# try n_neighbors from 1 to 10
neighbors_settings = range(1, 11)

for n_neighbors in neighbors_settings:
    # build the model
    clf = KNeighborsClassifier(n_neighbors=n_neighbors)
    clf.fit(X_train, y_train)
    # record training set accuracy
    training_accuracy.append(clf.score(X_train, y_train))
    # record generalization accuracy
    test_accuracy.append(clf.score(X_test, y_test))

plt.plot(neighbors_settings, training_accuracy, label="training accuracy")
plt.plot(neighbors_settings, test_accuracy, label="test accuracy")
plt.ylabel("Accuracy")
plt.xlabel("n_neighbors")
plt.legend()
```

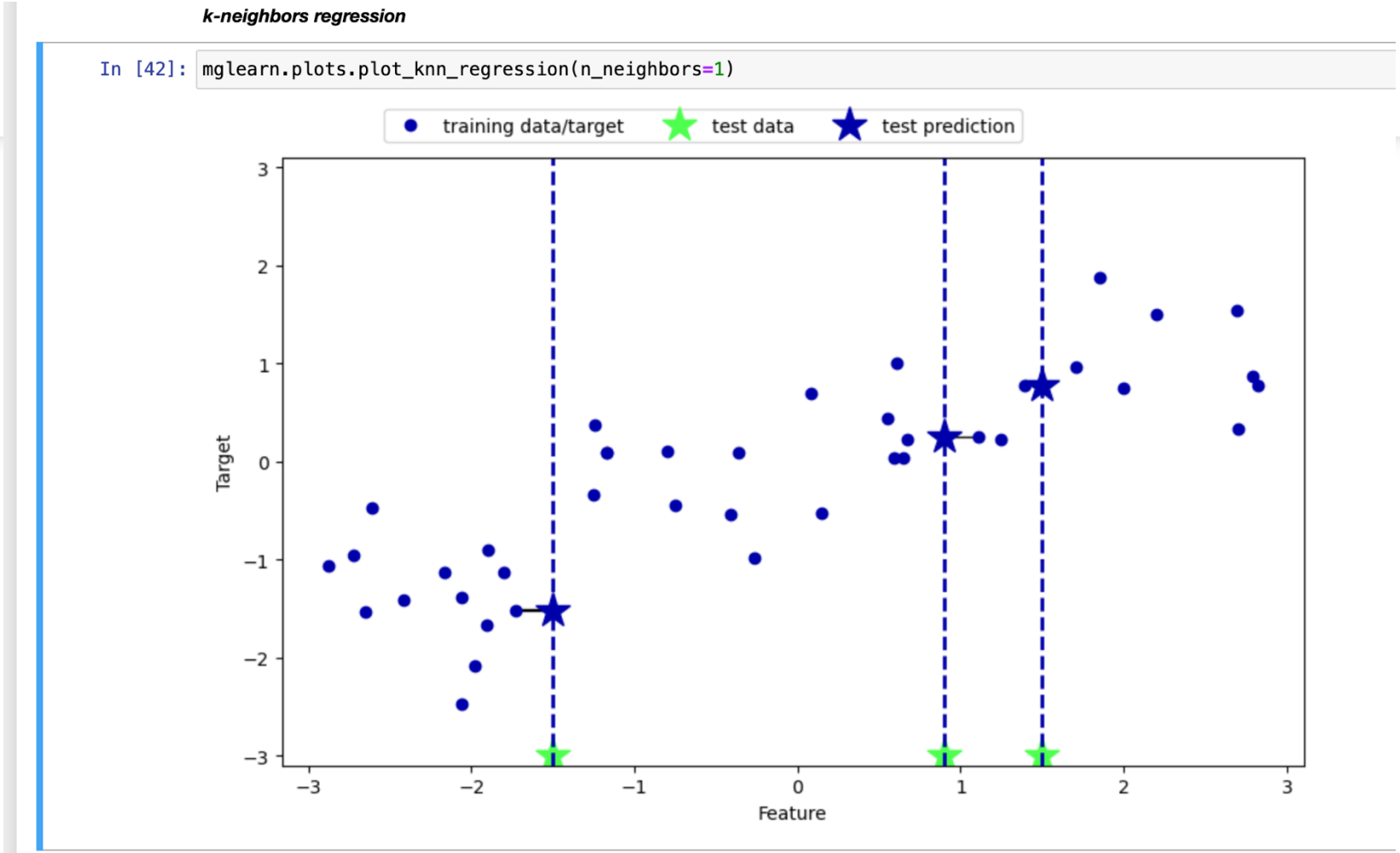
Some analysis from the previous plot

The plot shows the training and test set accuracy on the y-axis against the setting of n_neighbors on the x-axis. While real-world plots are rarely very smooth, we can still recognize some of the characteristics of overfitting and underfitting (note that because considering fewer neighbors corresponds to a more complex model, the plot is horizontally flipped relative to the illustration in model complexity figure presented earlier). Considering a single nearest neighbor, the prediction on the training set is perfect. But when more neighbors are considered, the model becomes simpler and the training accuracy drops. The test set accuracy for using a single neighbor is lower than when using more neighbors, indicating that using the single nearest neighbor leads to a model that is too complex. On the other hand, when considering 10 neighbors, the model is too simple and performance is even worse. The best performance is somewhere in the middle, using around six neighbors. Still, it is good to keep the scale of the plot in mind. The worst performance is around 88% accuracy, which might still be acceptable.

k-neighbors regression

This is a regression variant of the k-nearest neighbors algorithm

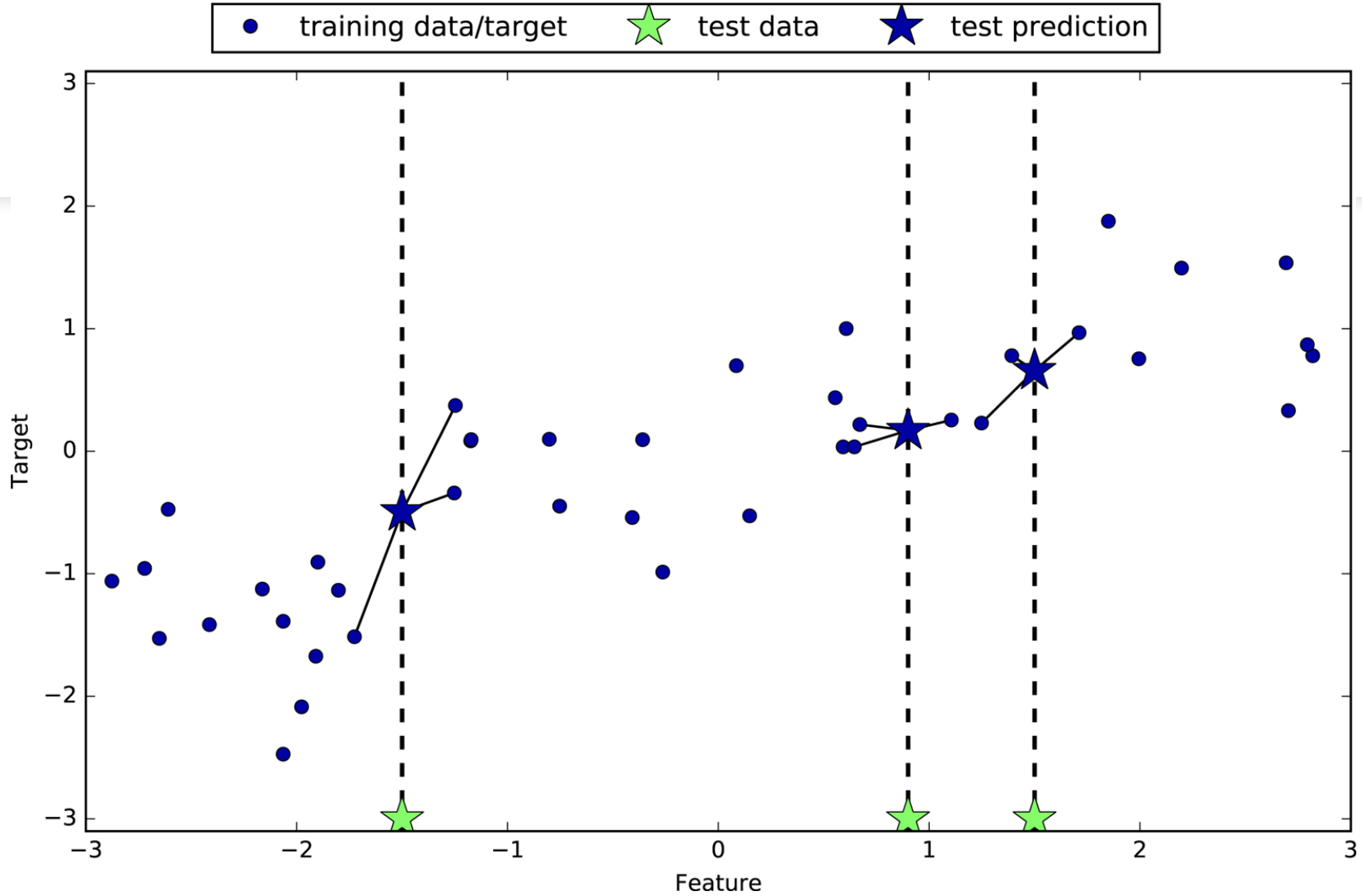
Using the single nearest neighbor, this time using the wave dataset. We've added three test data points as green stars on the x-axis. The prediction using a single neighbor is just the target value of the nearest neighbor. These are shown as blue stars in the Figure



Predictions made by one-nearest-neighbor regression on the wave dataset

we can use more than the single closest neighbor for regression. When using multiple nearest neighbors, the prediction is the average, or mean, of the relevant neighbors:

```
mglearn.plots.plot_knn_regression(n_neighbors=3)
```



Predictions made by three-nearest-neighbors regression on the wave dataset

The k-nearest neighbors algorithm for regression is implemented in the KneighborsRegressor class in scikit-learn. It's used similarly toKNeighborsClassifier:

In [46]:
```
from sklearn.neighbors import KNeighborsRegressor

X, y = mglearn.datasets.make_wave(n_samples=40)

# split the wave dataset into a training and a test set
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=0)

# instantiate the model and set the number of neighbors to consider to 3
reg = KNeighborsRegressor(n_neighbors=3)
# fit the model using the training data and training targets
reg.fit(X_train, y_train)
```

Out[46]:
```
KNeighborsRegressor
KNeighborsRegressor(n_neighbors=3)
```

Test set predictions:

```
In [47]: print("Test set predictions:\n", reg.predict(X_test))

Test set predictions:
 [-0.054  0.357  1.137 -1.894 -1.139 -1.631  0.357  0.912 -0.447 -1.139]
```

R Squared Evaluation:

We can also evaluate the model using the score method, which for regressors returns the R2 score. The R2 score, also known as the coefficient of determination, is a measure of goodness of a prediction for a regression model, and yields a score between 0 and 1. A value of 1 corresponds to a perfect prediction, and a value of 0 corresponds to a constant model that just predicts the mean of the training set responses, y_train

```
In [48]: print("Test set R^2: {:.2f}".format(reg.score(X_test, y_test)))

Test set R^2: 0.83
```

the score is 0.83, which indicates a relatively good model fit.

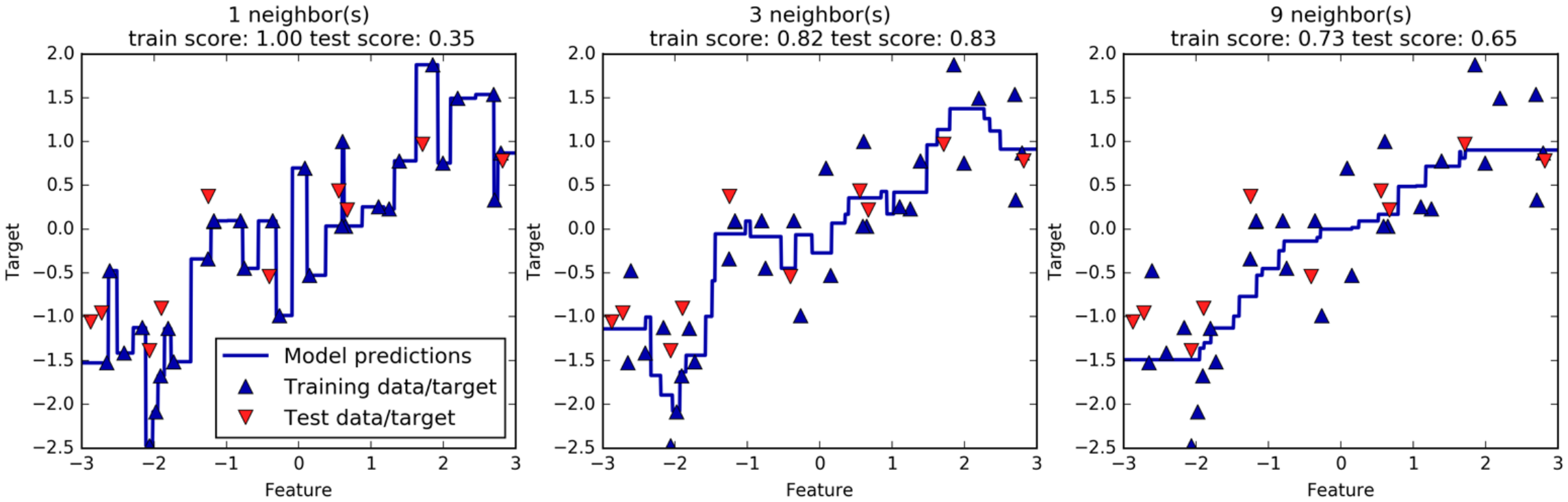
Analyzing KNeighborsRegressor

For our one-dimensional dataset, we can see what the predictions look like for all possible feature values (Figure bellow). To do this, we create a test dataset consisting of many points on the line:

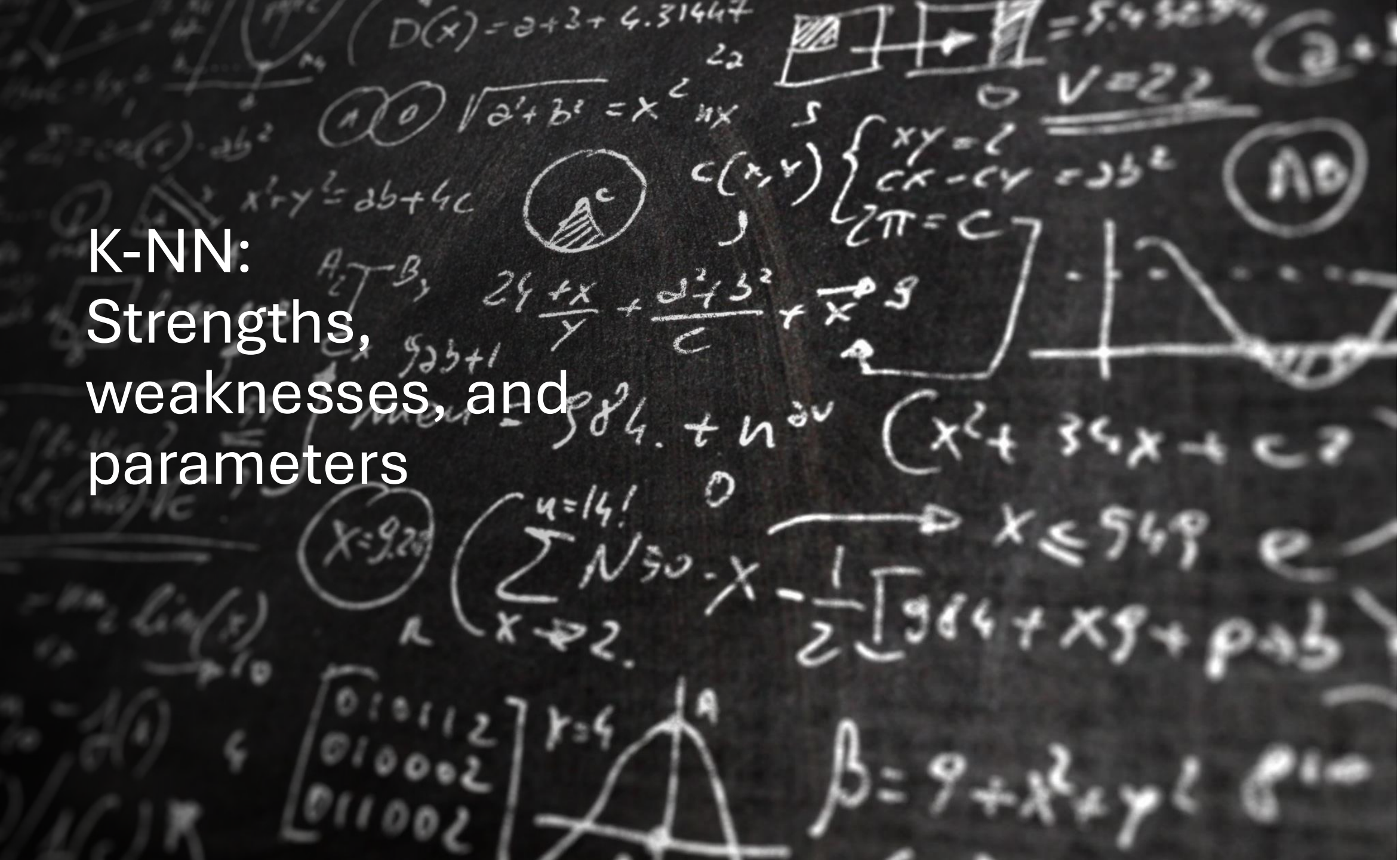
In [49]:
```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
# create 1,000 data points, evenly spaced between -3 and 3
line = np.linspace(-3, 3, 1000).reshape(-1, 1)
for n_neighbors, ax in zip([1, 3, 9], axes):
    # make predictions using 1, 3, or 9 neighbors
    reg = KNeighborsRegressor(n_neighbors=n_neighbors)
    reg.fit(X_train, y_train)
    ax.plot(line, reg.predict(line))
    ax.plot(X_train, y_train, '^', c=mglearn.cm2(0), markersize=8)
    ax.plot(X_test, y_test, 'v', c=mglearn.cm2(1), markersize=8)

    ax.set_title(
        "{} neighbor(s)\n train score: {:.2f} test score: {:.2f}".format(
            n_neighbors, reg.score(X_train, y_train),
            reg.score(X_test, y_test)))
    ax.set_xlabel("Feature")
    ax.set_ylabel("Target")
axes[0].legend(["Model predictions", "Training data/target",
                "Test data/target"], loc="best")
```

Out[49]:
```
<matplotlib.legend.Legend at 0x15a168150>
```

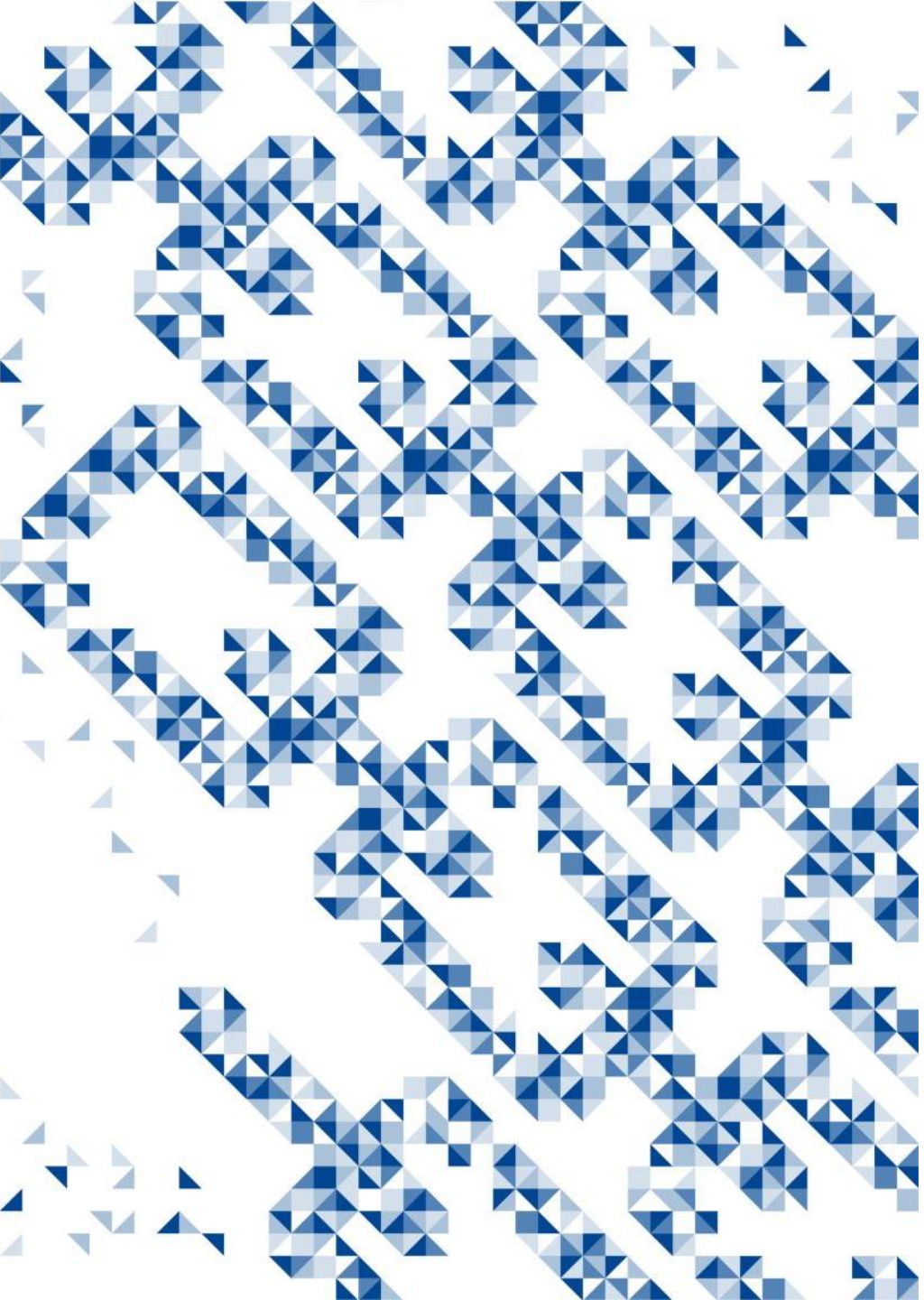
Comparing predictions made by nearest neighbors regression for different values of n_neighbors (code in the previous slide)



As we can see from the plot, using only a single neighbor, each point in the training set has an obvious influence on the predictions, and the predicted values go through all of the data points. This leads to a very unsteady prediction. Considering more neighbors leads to smoother predictions, but these do not fit the training data as well.



K-NN: Strengths, weaknesses, and parameters



Parameters

In principle, there are two important parameters to the KNeighbors classifier:

1. The number of neighbors: In practice, using a small number of neighbors like three or five often works well, but you should certainly adjust this parameter when required.
2. How you measure distance between data points (Typically Euclidian distance which works in many settings but there are others as well)

K-NN Strengths

k-NN model is very easy to understand

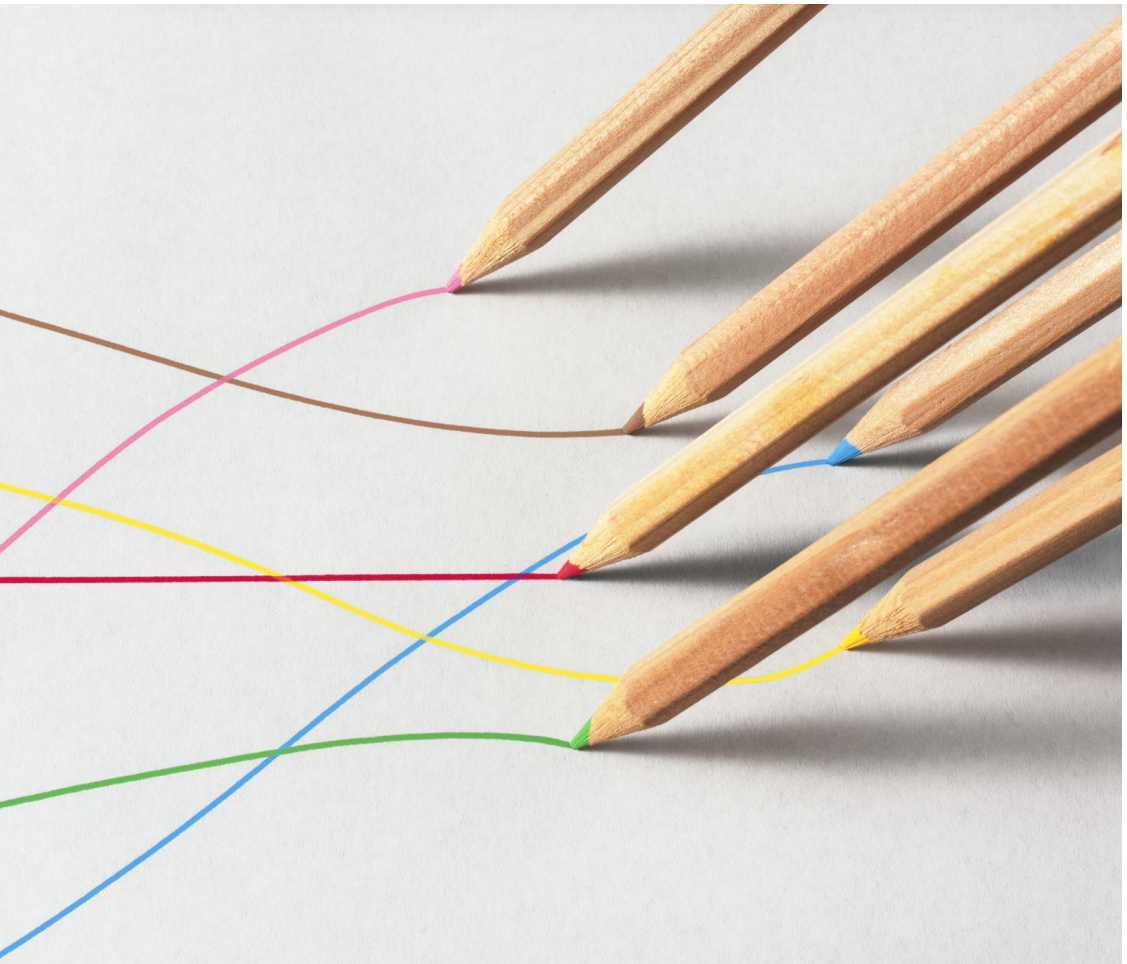
Often gives reasonable performance without a lot of adjustments.

K-NN Weaknesses

Building the nearest neighbors model is usually very fast, but when your training set is very large (either in number of features or in number of samples) prediction can be slow

This approach often does not perform well on datasets with many features (hundreds or more), and it does particularly badly with datasets where most features are 0 most of the time (so-called sparse datasets).

Finally on k-NN



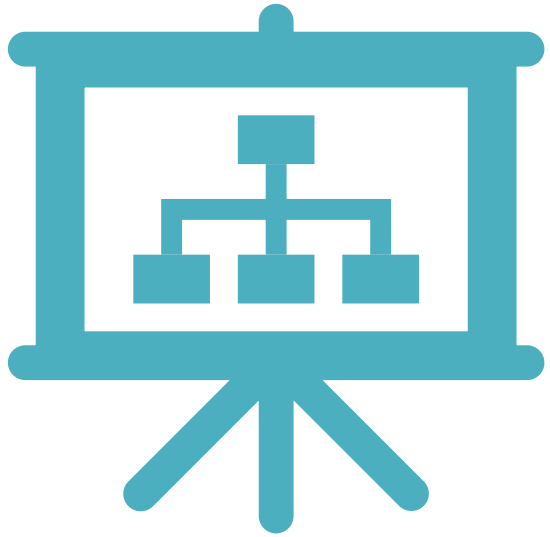
while the nearest k-neighbors algorithm is easy to understand, it is not often used in practice, due to prediction being slow and its inability to handle many features. The method we discuss next (Linear Models for Regression and Classification) has neither of these drawbacks.

Linear Models

Outline:

1. Linear models for regression: Linear regression (aka ordinary least squares), Ridge regression, Lasso,

2. Linear models for classification: LogisticRegression, linear support vector machines



Linear Models

Linear models are a class of models that are widely used in practice and have been studied extensively in the last few decades, with roots going back over a hundred years. Linear models make a prediction using a linear function of the input features.

Linear models for regression

For regression, the general prediction formula for a linear model looks as follows:

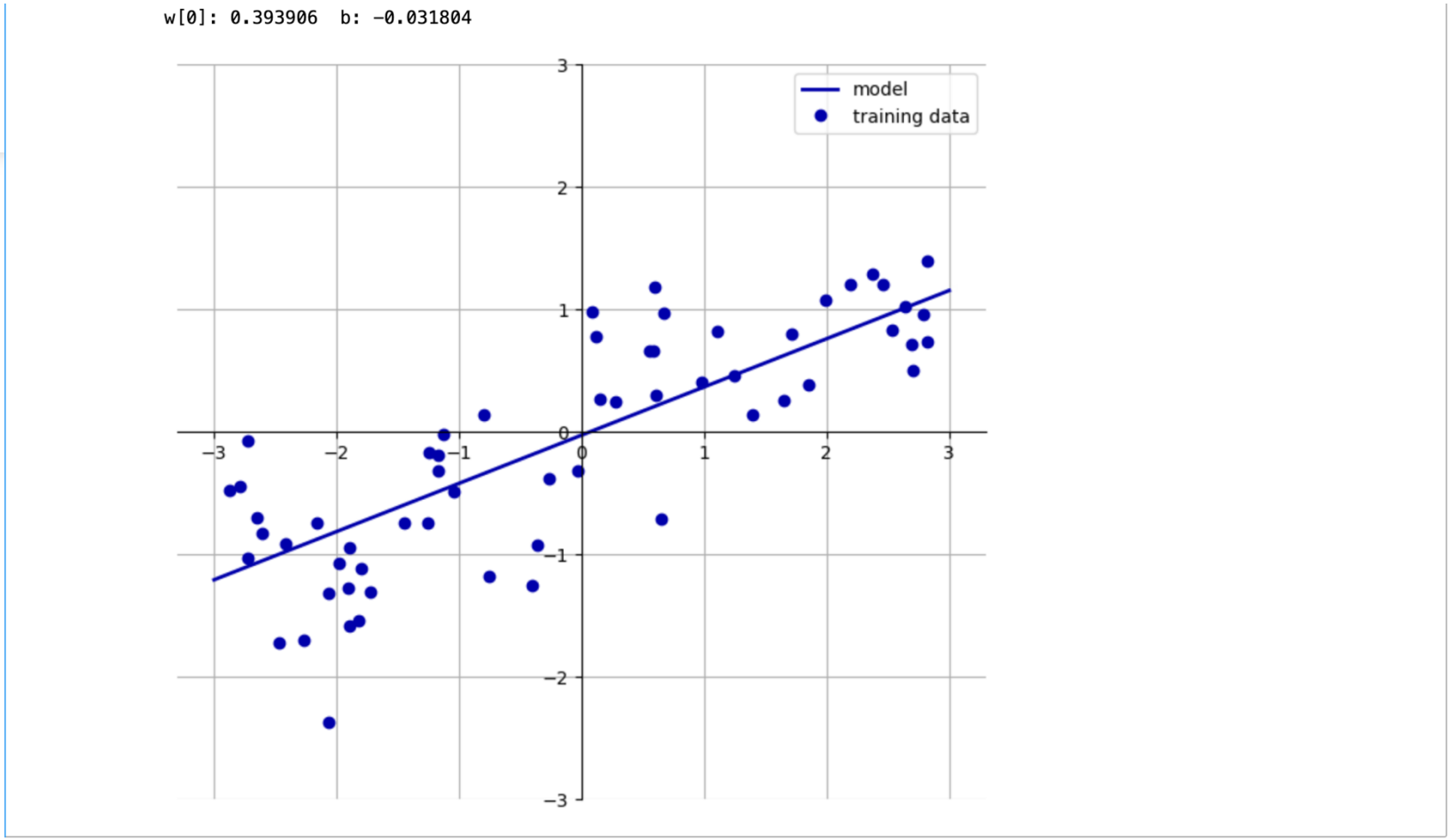
$\hat{y} = w[0] * x[0] + w[1] * x[1] + \ldots + w[p] * x[p] + b$

Here, x[0] to x[p] denotes the features (in this example, the number of features is p) of a single data point, w and b are parameters of the model that are learned, and $\hat{y}$ is the prediction the model makes. For a dataset with a single feature, this is:

$\hat{y} = w[0] * x[0] + b$

which you might remember from high school mathematics as the equation for a line. Here, w[0] is the slope and b is the y-axis offset. For more features, w contains the slopes along each feature axis. Alternatively, you can think of the predicted response as being a weighted sum of the input features, with weights (which can be negative) given by the entries of w.

Let's look back on the provided one dimensional wave dataset: *mglearn.plots.plot_linear_regression_wave()*



Predictions of a linear model on the wave dataset

Linear Models: Intro

We added a coordinate cross into the plot to make it easier to understand the line. Looking at w[0] we see that the slope should be around 0.4, which we can confirm visually in the plot. The intercept is where the prediction line should cross the y-axis: this is slightly below zero, which you can also confirm in the image. Linear models for regression can be characterized as regression models for which the prediction is a line for a single feature, a plane when using two features, or a hyperplane in higher dimensions (that is, when using more features).

Linear Models: Intro

If you compare the predictions made by the straight line with those made by the KNeighborsRegressor, using a straight line to make predictions seems very restrictive. It looks like all the fine details of the data are lost. In a sense, this is true. It is a strong (and somewhat unrealistic) assumption that our target y is a linear. combination of the features. But looking at one-dimensional data gives a somewhat skewed perspective. For datasets with many features, linear models can be very powerful. In particular, if you have more features than training data points, any target y can be perfectly modeled (on the training set) as a linear function

Note

There are many different linear models for regression. The difference between these models lies in how the model parameters w and b are learned from the training data, and how model complexity can be controlled.

Linear regression (aka ordinary least squares)

Linear regression, or ordinary least squares (OLS), is the simplest and most classic linear method for regression. Linear regression finds the parameters w and b that minimize the mean squared error between predictions and the true regression targets, y, on the training set. The mean squared error is the sum of the squared differences between the predictions and the true values. Linear regression has no parameters, which is a benefit, but it also has no way to control model complexity.

```
from sklearn.linear_model import LinearRegression

X, y = mglearn.datasets.make_wave(n_samples=60)

X_train, X_test, y_train, y_test = train_test_split(X, y,
random_state=42)

lr = LinearRegression().fit(X_train, y_train)
```

The "slope" parameters (w), also called weights or coefficients, are stored in the `coef_` attribute, while the offset or intercept (b) is stored in the `intercept_` attribute:

```
print("lr.coef_: {}".format(lr.coef_))
print("lr.intercept_: {}".format(lr.intercept_))
```

You might notice the strange-looking trailing underscore at the end of coef_ and intercept_. scikit-learn always stores anything that is derived from the training data in attributes that end with a trailing underscore. That is to separate them from parameters that are set by the user.

Linear Regression

The intercept_ attribute is always a single float number, while the coef_ attribute is a NumPy array with one entry per input feature. As we only have a single input feature in the wave dataset, lr.coef_ only has a single entry. Let's look at the training set and test set performance:

```
print("Training set score: {:.2f}".format(lr.score(X_train,
y_train)))
print("Test set score: {:.2f}".format(lr.score(X_test, y_test)))
```

R-Squared and Linear Regression Overfitting or Underfitting from our dataset/results?

An R2 of around 0.66 is not very good, but we can see that the scores on the training and test sets are very close together. This means we are likely underfitting, not overfitting. For this one-dimensional dataset, there is little danger of overfitting, as the model is very simple (or restricted). However, with higher-dimensional datasets (meaning datasets with a large number of features), linear models become more powerful, and there is a higher chance of overfitting.

Let’s explore the California housing dataset…